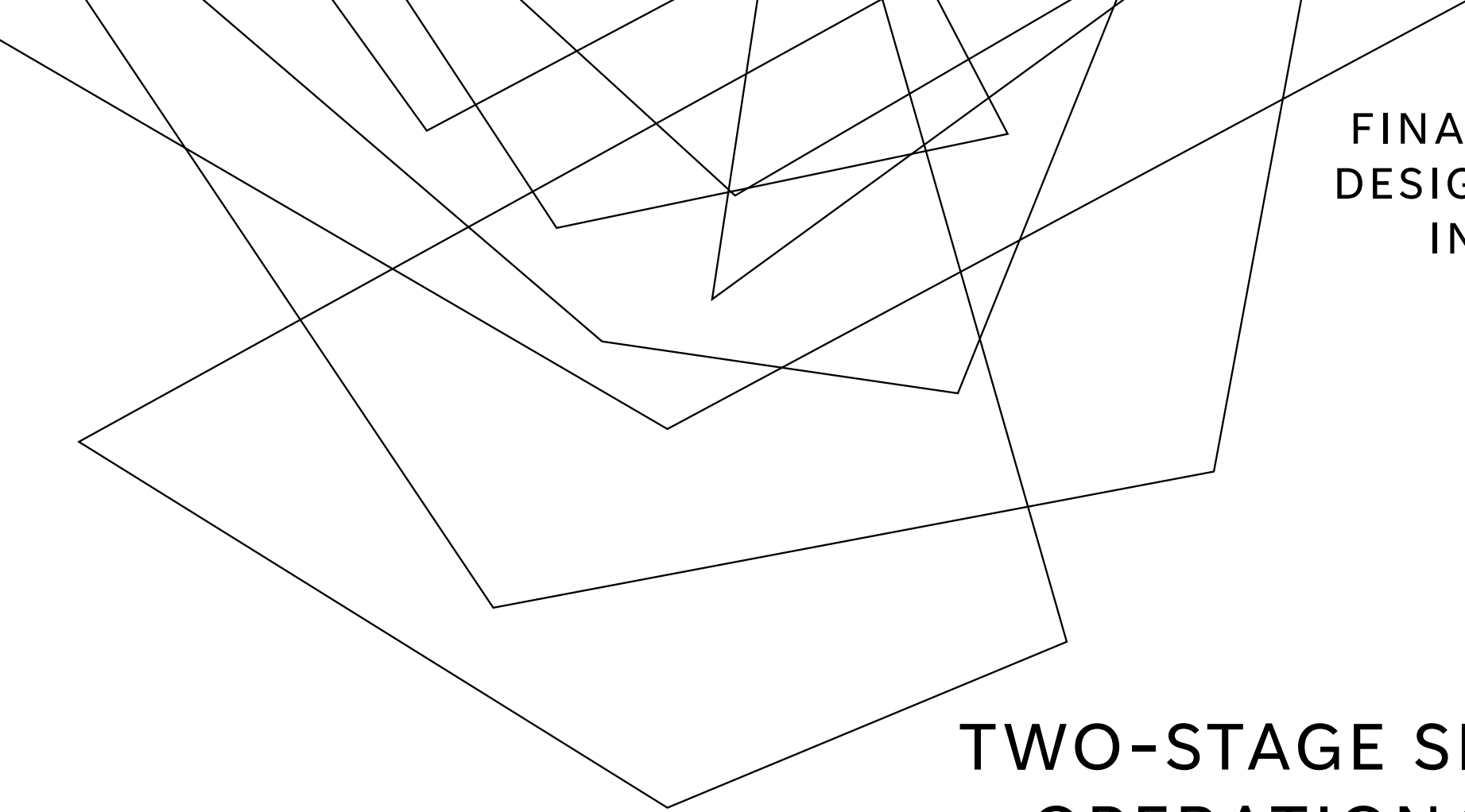


Abstract geometric lines in black on a white background, forming various overlapping polygons and shapes, primarily located on the left side of the page.

FINAL PROJECT 01205345
DESIGN OF CMOS ANALOG
INTEGRATED CIRCUITS

TWO-STAGE SINGLE-ENDED OPERATIONAL AMPLIFIER

Wiwitthawin Chareonngam 6310501933

CONTENT

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DESIGN SPECIFICATION

1. DC gain ≥ 80 dB
2. Output swing ≥ 1.5 V
3. Unity-gain frequency ≥ 7 MHz
4. Phase margin of OL transfer function ≥ 60 degrees
5. ICMR ≥ 1 V
6. Total integrated input-referred noise (1Hz -> 1GHz) $\leq 220 \mu V_{rms}$
7. 1% settling time $< 2 \mu s$ with 0.5V step input

8. THD of $V_{out} \leq 0.2 \%$ when $V_{out} = 1.5 V_{pp}$

* 0.18 μm CMOS process, $V_{DD} = 1.8$ V, $C_L = 100$ fF,

1 ideal current source = 1 μA , $P \leq 6 \mu W$

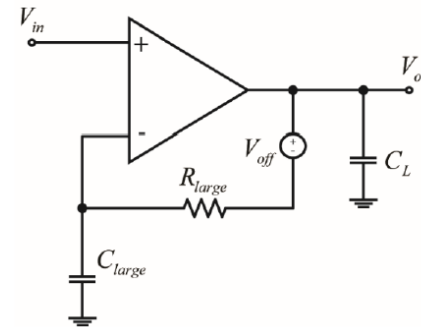


Figure 1

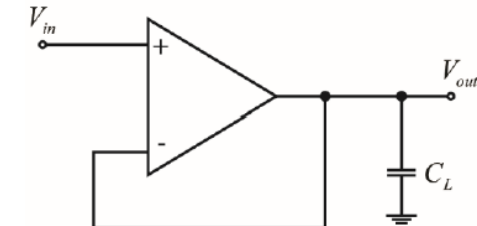


Figure 2

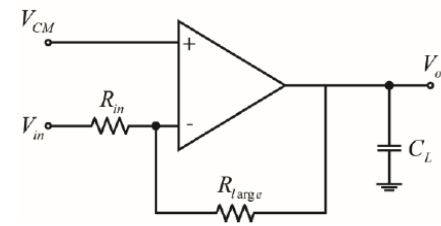
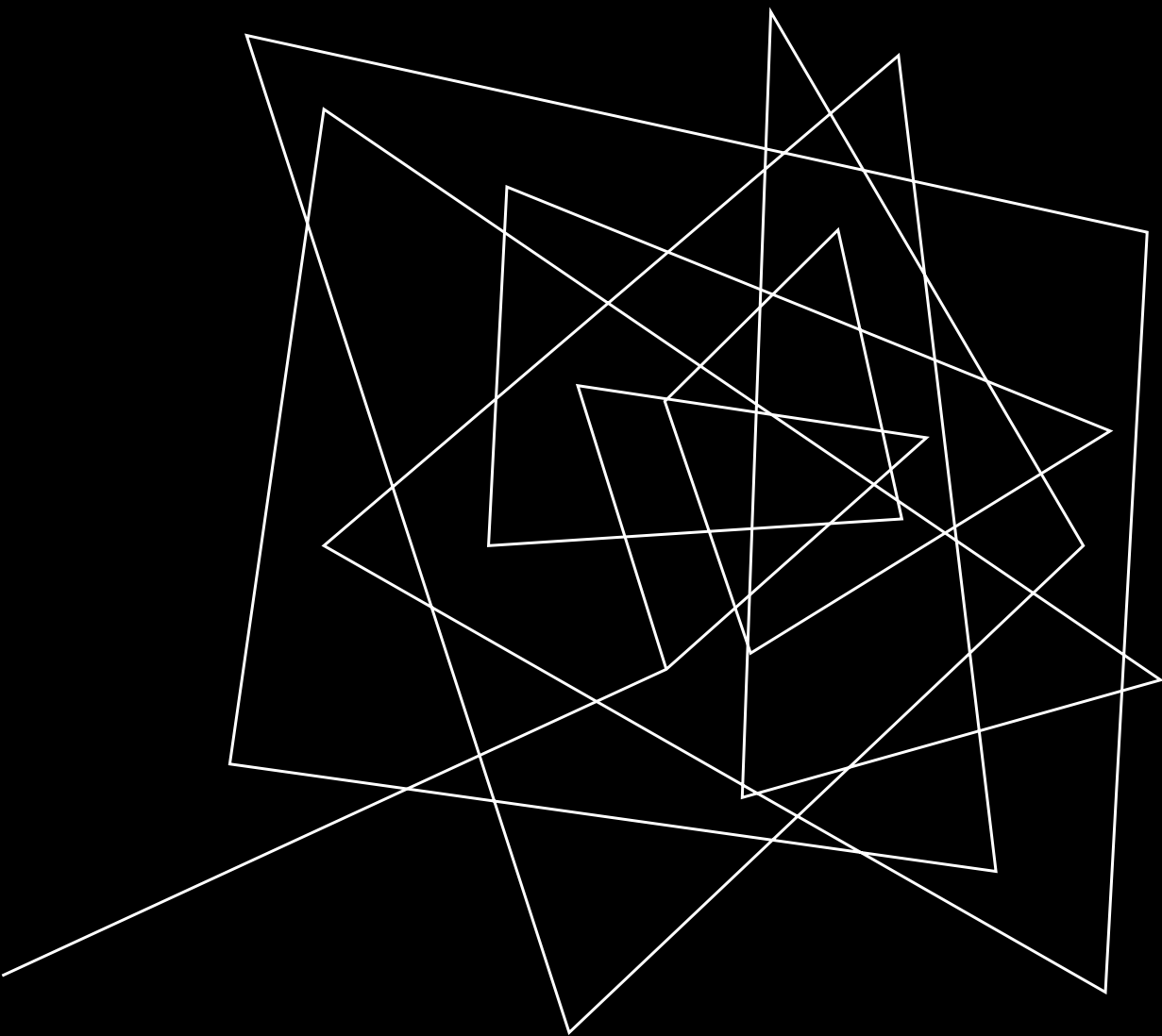
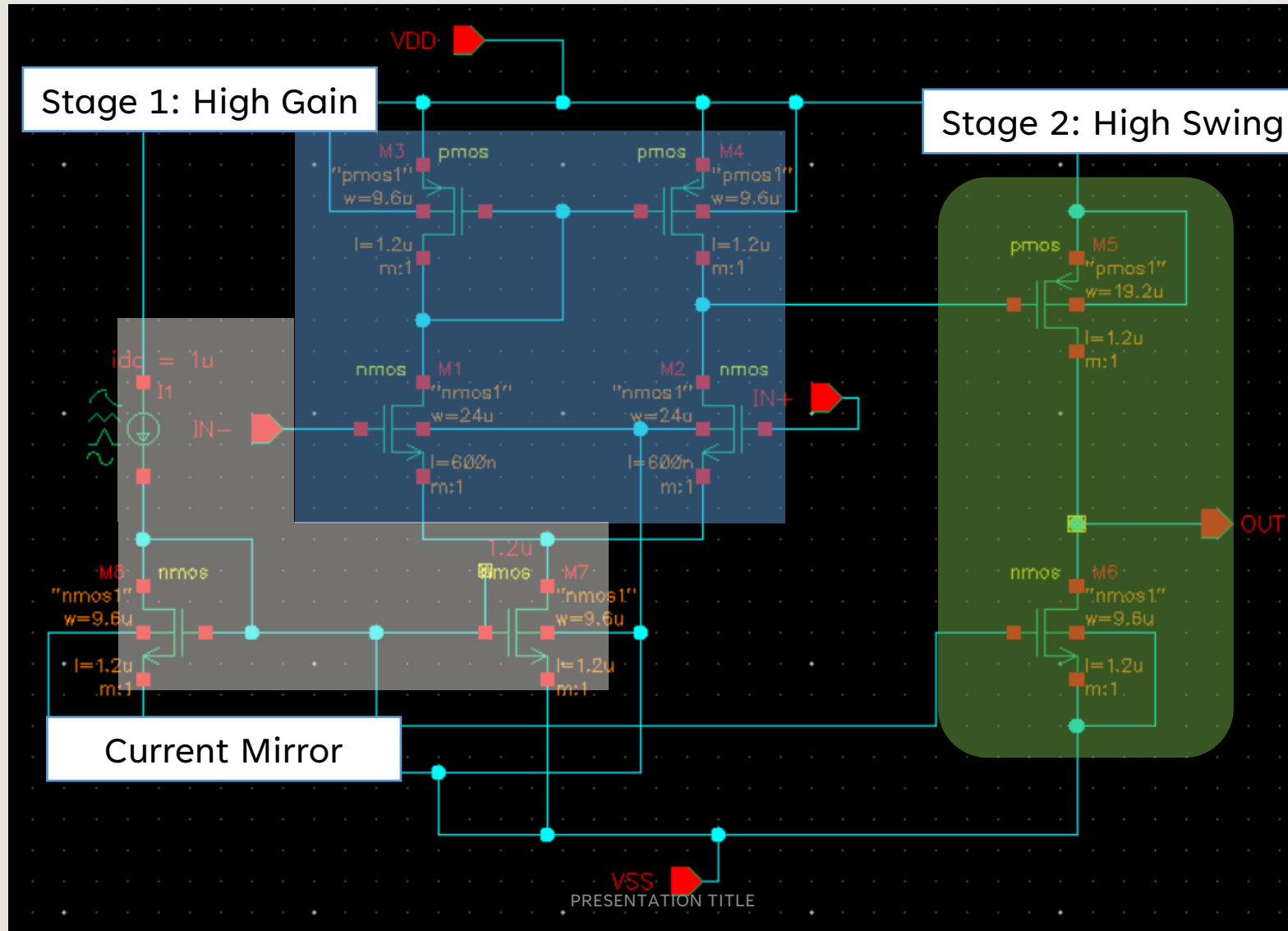


Figure 3.



METHODOLOGY

UNCOMPENSATED TWO-STAGE SINGLE-ENDED OPERATIONAL AMPLIFIER



DC BIAS OF UNCOMPENSATED CIRCUIT

From $P \leq 6 \mu W \rightarrow$ Assume Each Stage consume at least $1 \mu A$ and

$$\text{from that } \left(\frac{W}{L}\right)_3 = \left(\frac{W}{L}\right)_4 = \left(\frac{W}{L}\right)_b = \left(\frac{W}{L}\right)_6 = \frac{1}{2} \left(\frac{W}{L}\right)_5$$

In M1&M2: use $Gm \gg I_D$ for better gain and better $1/f$ noise
(Increasing W and L also reducing $1/f$ noise but large L affects large settling time)

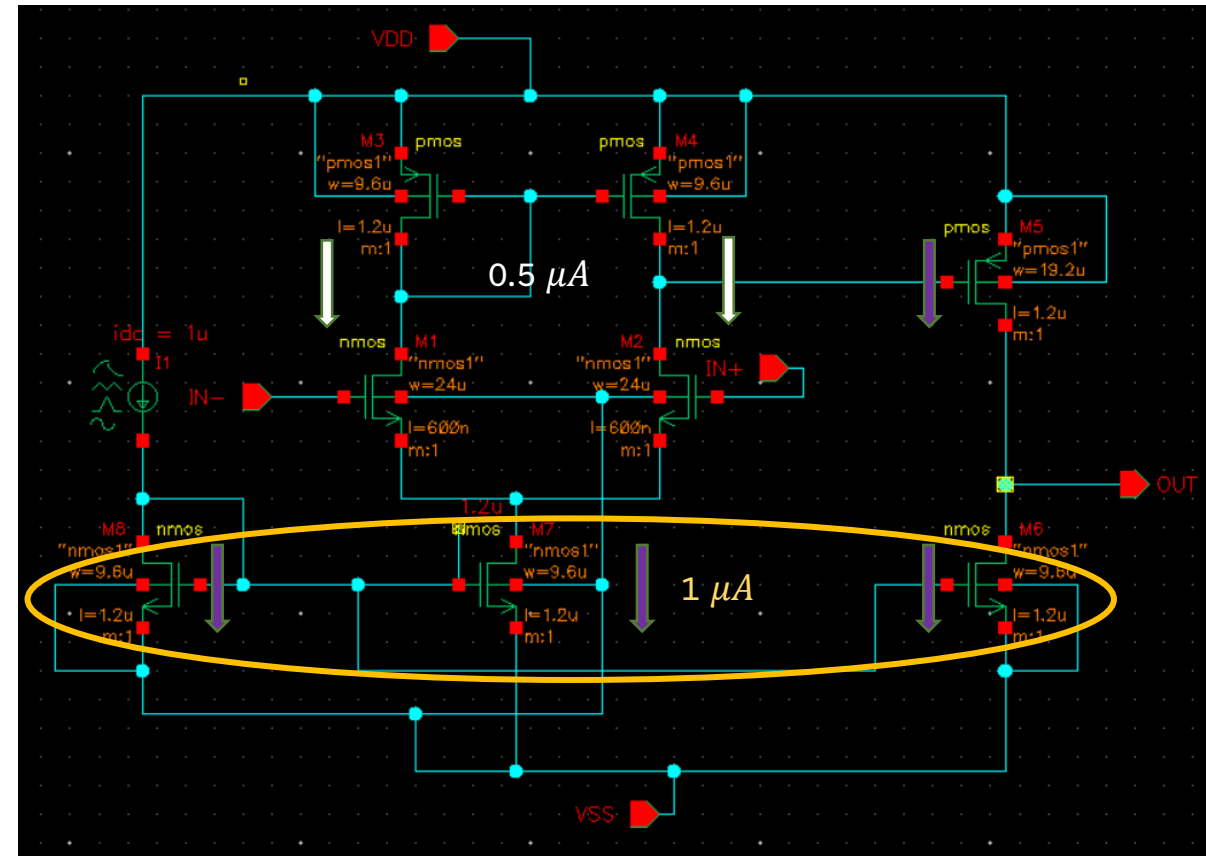
ICMR is bounded by $V_{ov3,4}$ and $V_{ov1,2,b} \rightarrow$

$$V_{GS1} + V_{ova,b} < V_{in,cm} < V_{DD} - |V_{SG3,4}| + V_{TH1}$$

If we decrease $\left(\frac{W}{L}\right)_{3,4}$, $V_{ov3,4}$, $V_{ova,b}$ and $|V_{SG3,4}|$ will increase

If we increase $\left(\frac{W}{L}\right)_{1,2}$, $V_{ov1,2}$ will decrease

For Large Output Swing: $V_{ov6} < V_{out} < V_{DD} - V_{ov5}$
 $\left(\frac{W}{L}\right)_{5,6}$ is to be increase to make $V_{ov5,6}$ smaller but large enough to make $M_{a,b}$ saturated $\rightarrow V_{GS} > V_{TH}$



DC GAIN OF COMPENSATED CIRCUIT

$$A_v = Gm_2(r_{o2} \parallel r_{o4}) * Gm_5(r_{o5} \parallel r_{o6})$$

$$Gm = \frac{2I_D}{V_{ov}}, \quad r_o = \frac{1}{\lambda I_D}, \quad \lambda = \frac{1}{L_{eff}} * \frac{\partial X_{dl}}{\partial V_{DS}}$$

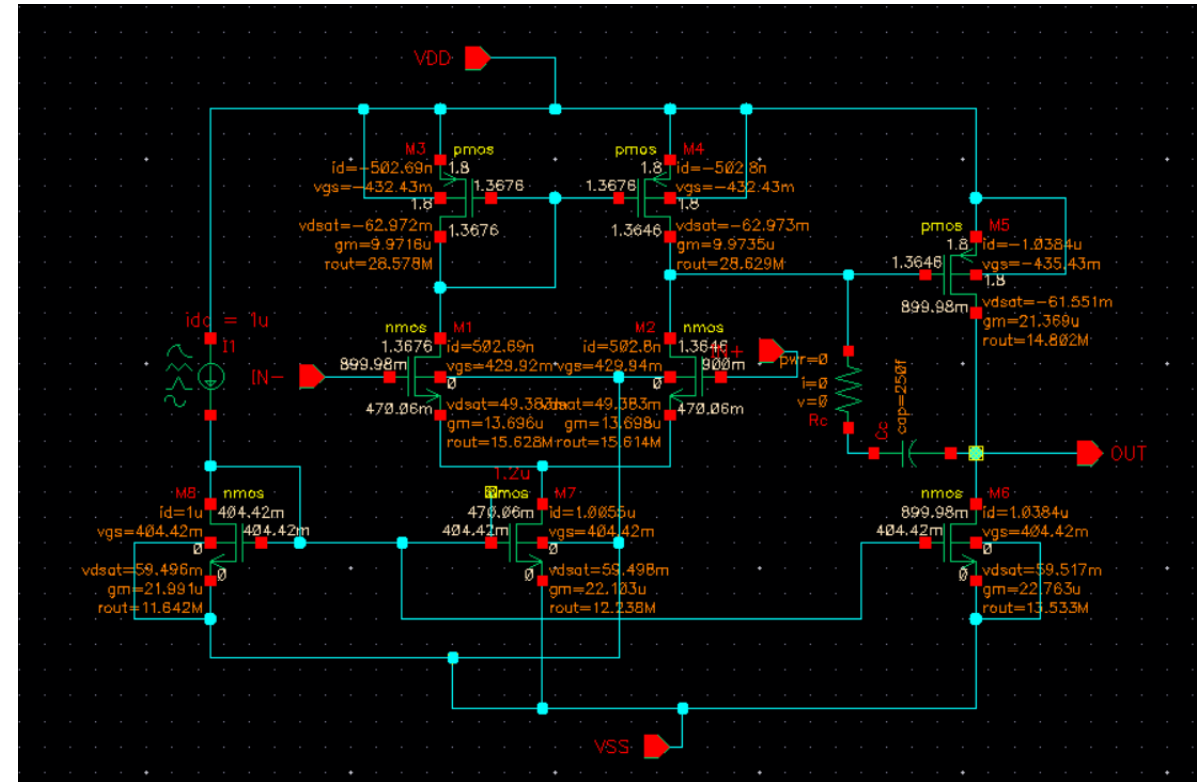
- For Stage1, we want high gain, so we need to maximize Gm and r_o but Gm can be increase by **reducing overdrive voltage** and r_o can be increase by reducing λ which result in **increasing L_{eff}**

For high gain

-> Increase $\frac{W}{L}$ of M1 and M2 to reduce V_{ov} and thus higher Gm (M1 and M2 operate in Subthreshold Conduction)

-> Increase L_{eff}

For Stage2, we don't need large gain, but we need large output swing which is from small overdrive voltage



UNITY GAIN AND PM OF COMPENSATED CIRCUIT

Design C_c and R_c of Compensation:

Unity Gain: $GBW = \frac{G_{m5}}{C_c * 2\pi}$ and $C_c \geq 0.22 * C_L$

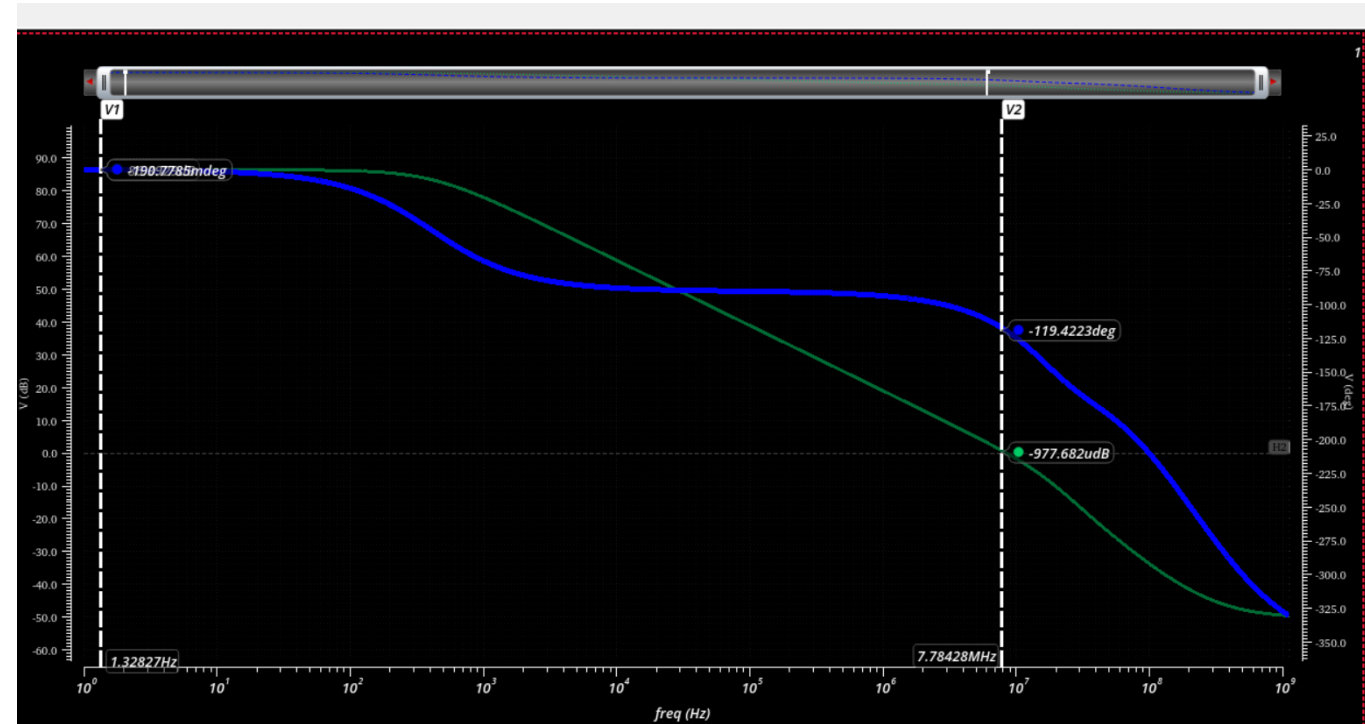
* $C_c \geq 22$ fF, $GBW = 7\text{MHz} \rightarrow C_c = 486$ fF

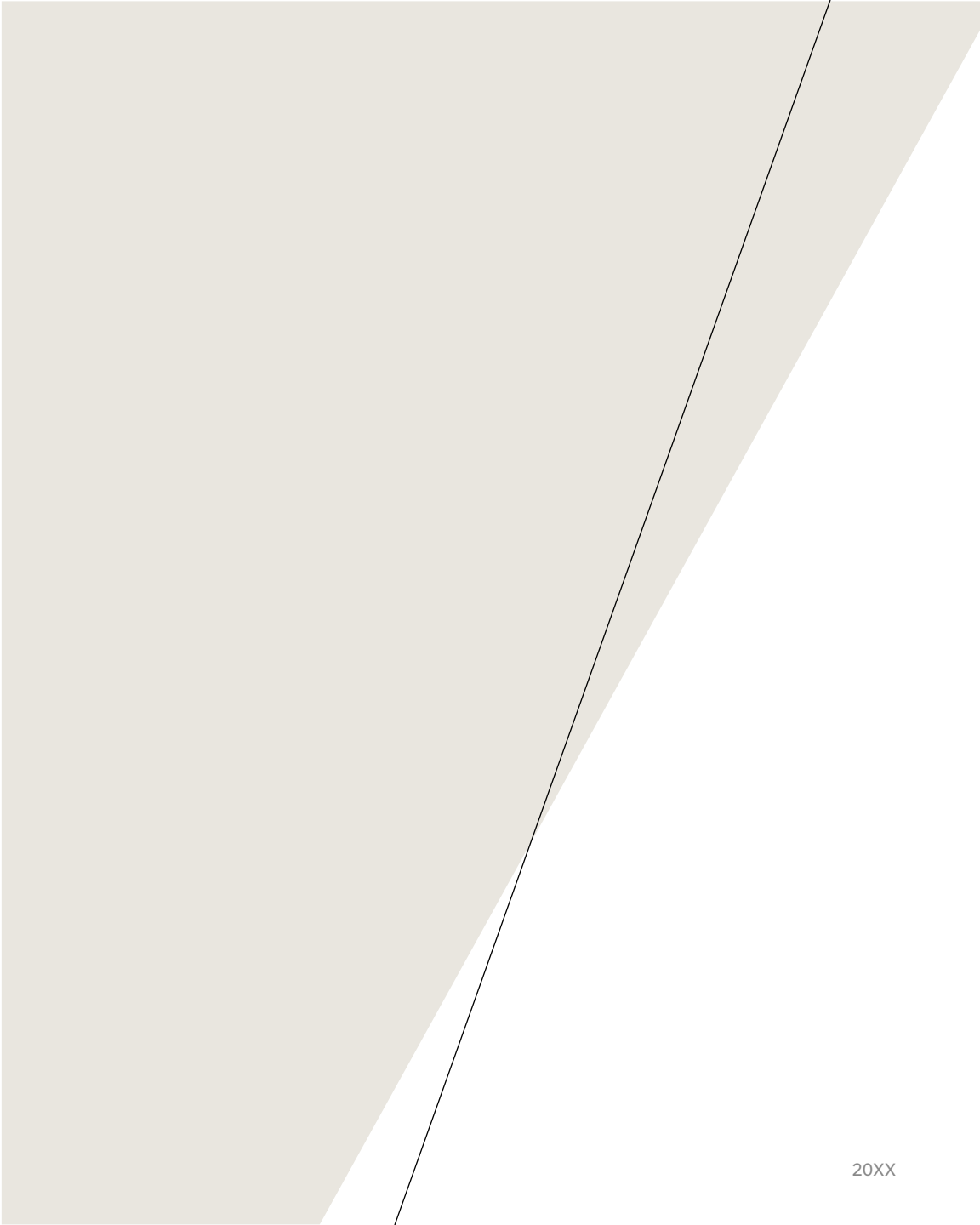
$PM = -\tan^{-1}\left(\frac{1}{10}\right) - \tan^{-1}(G_{m2}G_{m5}R_1R_2) - \tan^{-1}\left(\frac{GBW}{P_z}\right)$

$S_z = \frac{G_{m5}}{C_c} \frac{1}{1 - G_{m5}R_c}$; $G_{m5}R_c \leq 1$

$R_c \leq 46796$ k Ω , $C_c \geq 22$ fF, $C_c \approx 486$ fF

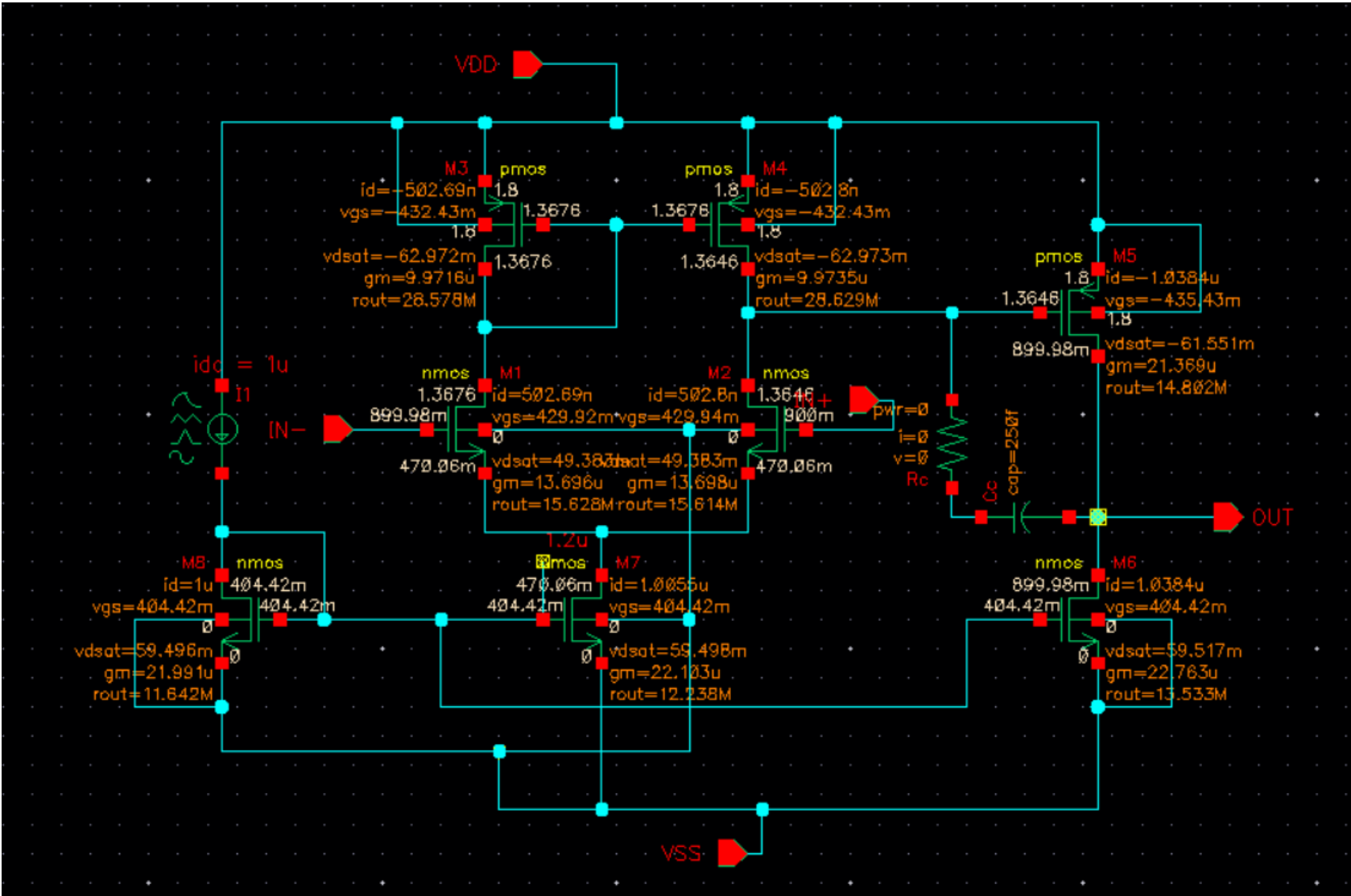
With greater C_c , comes with lower unity frequency and larger phase margin . R_c is to be far more than unity freq.





RESULTS

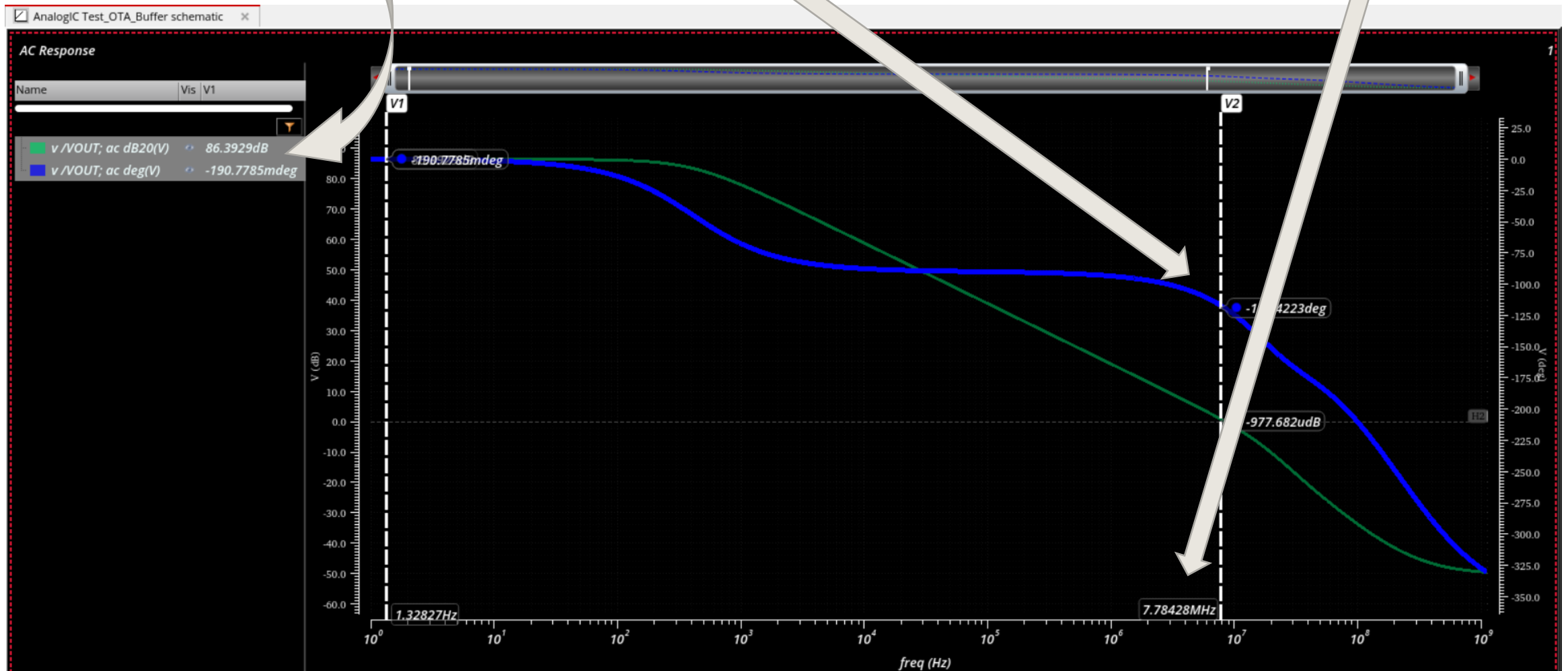
DC OPERATING POINT



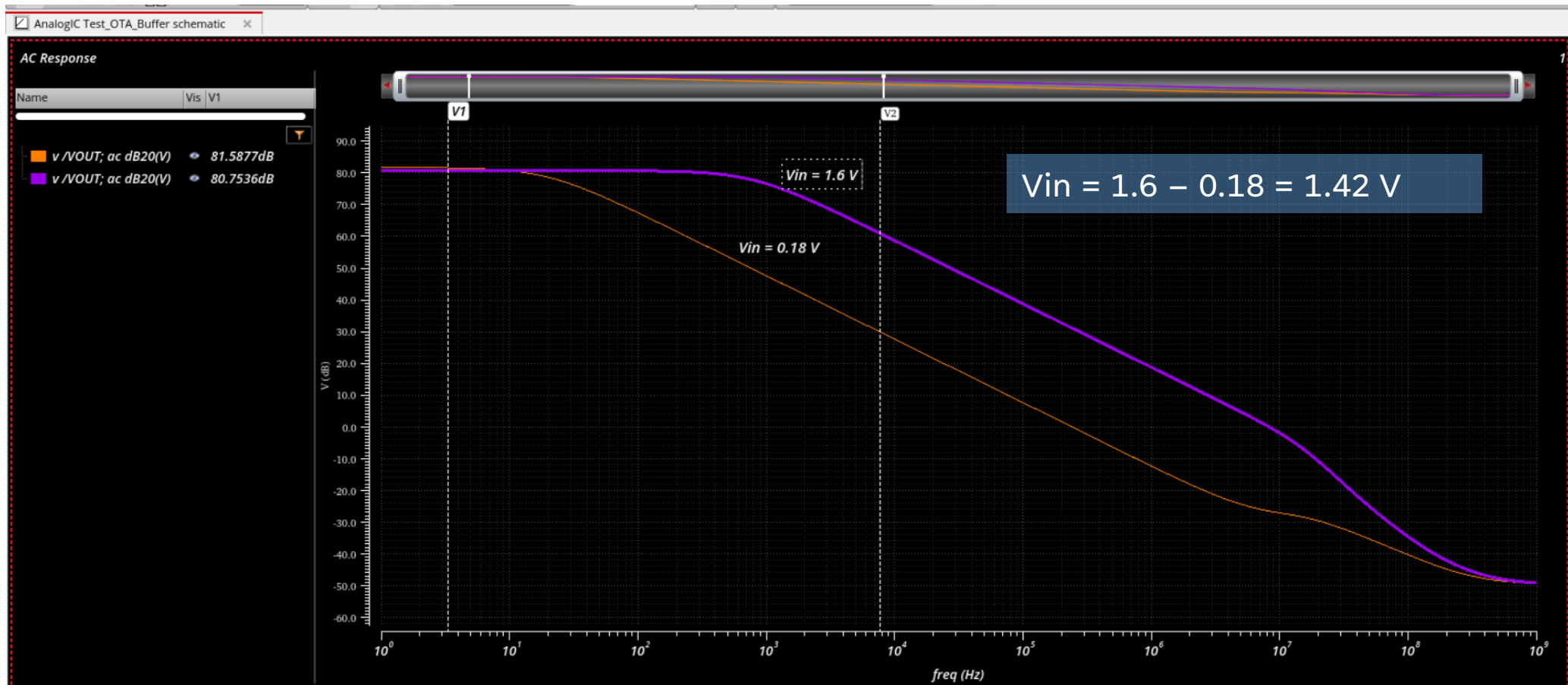
DC GAIN ≥ 80 DB

PHASE MARGIN ≥ 60 DEGREE

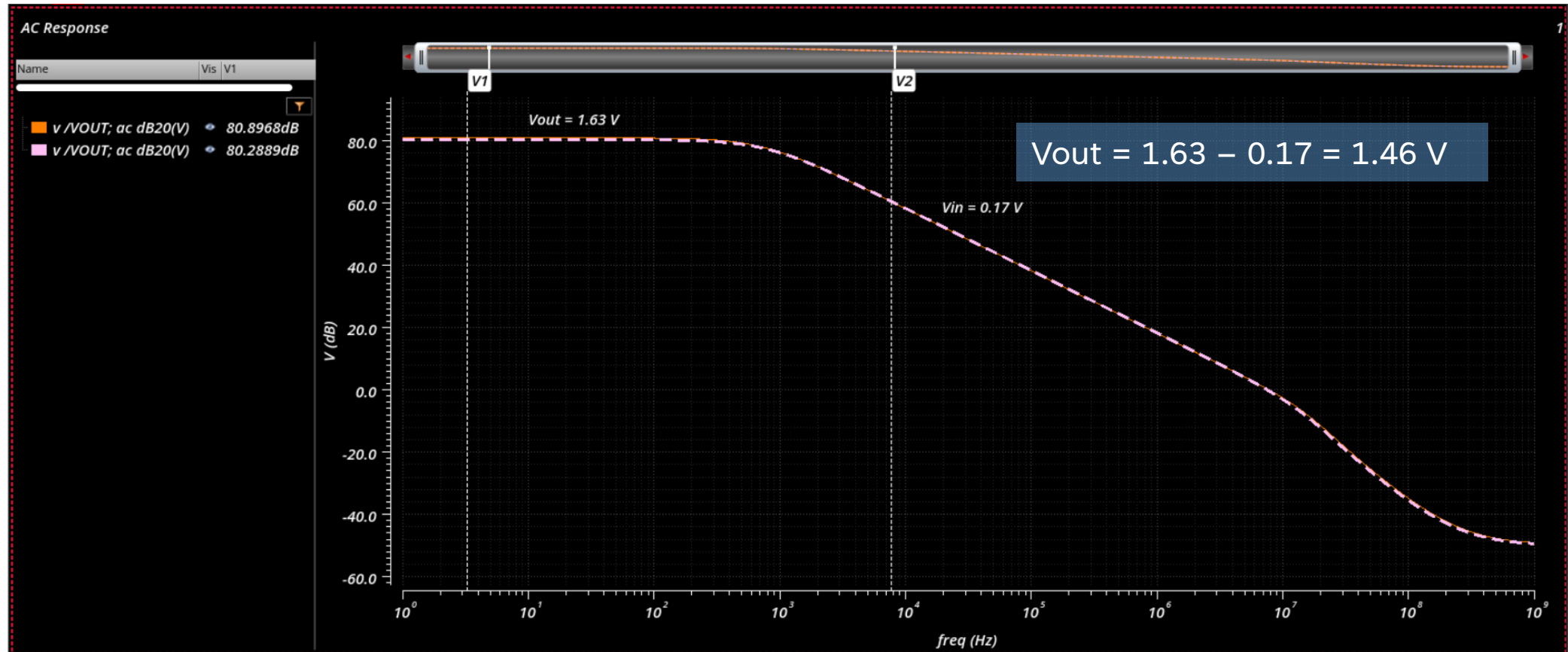
UNITY FREQUENCY ≥ 7 MHZ



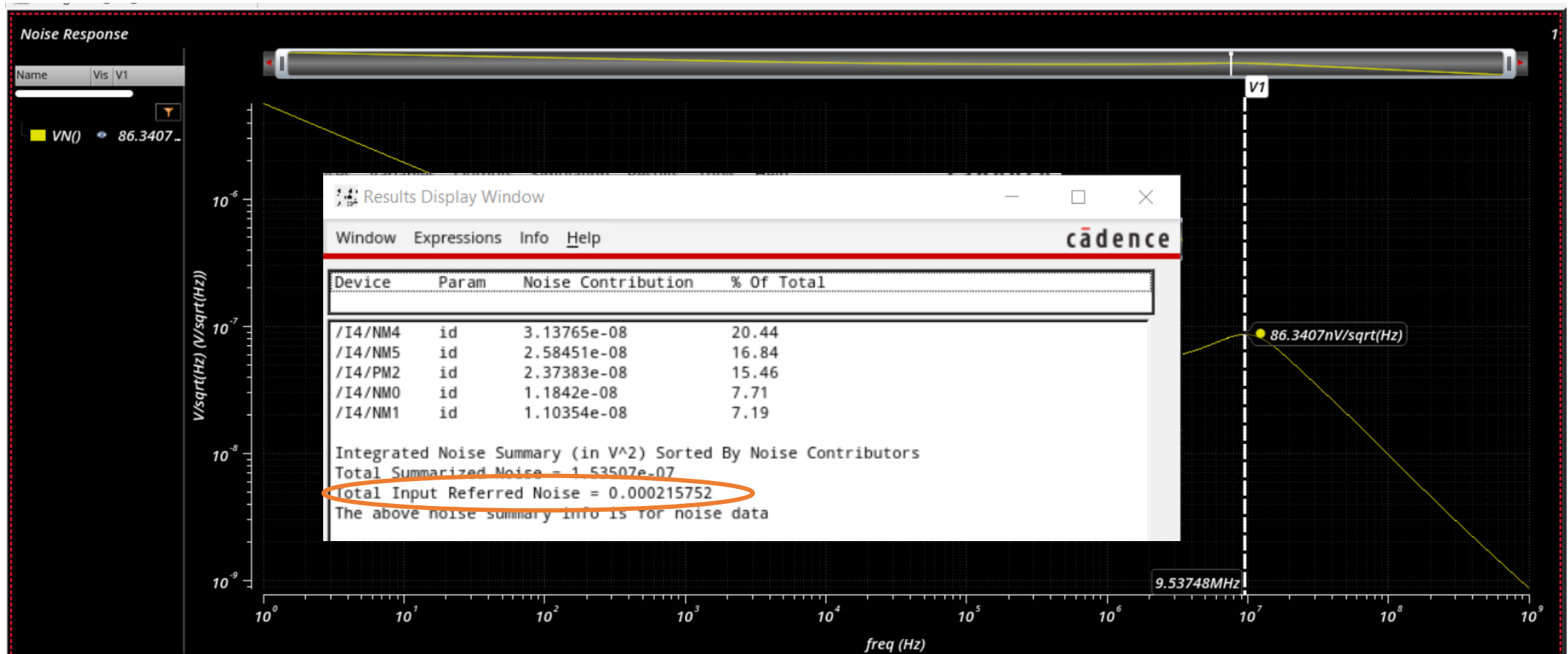
$$\text{ICMR} \geq 1\text{V}$$



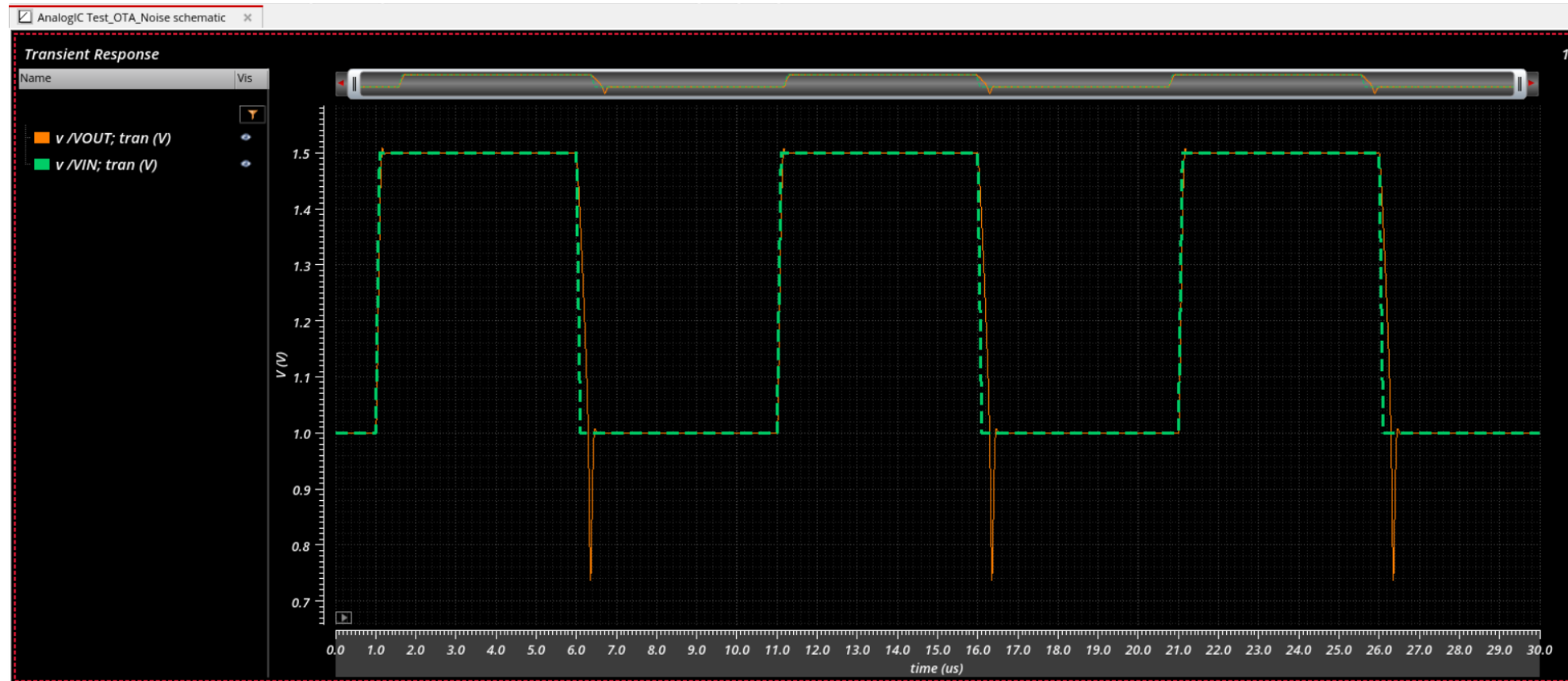
OUTPUT SWING ≥ 1.5 V



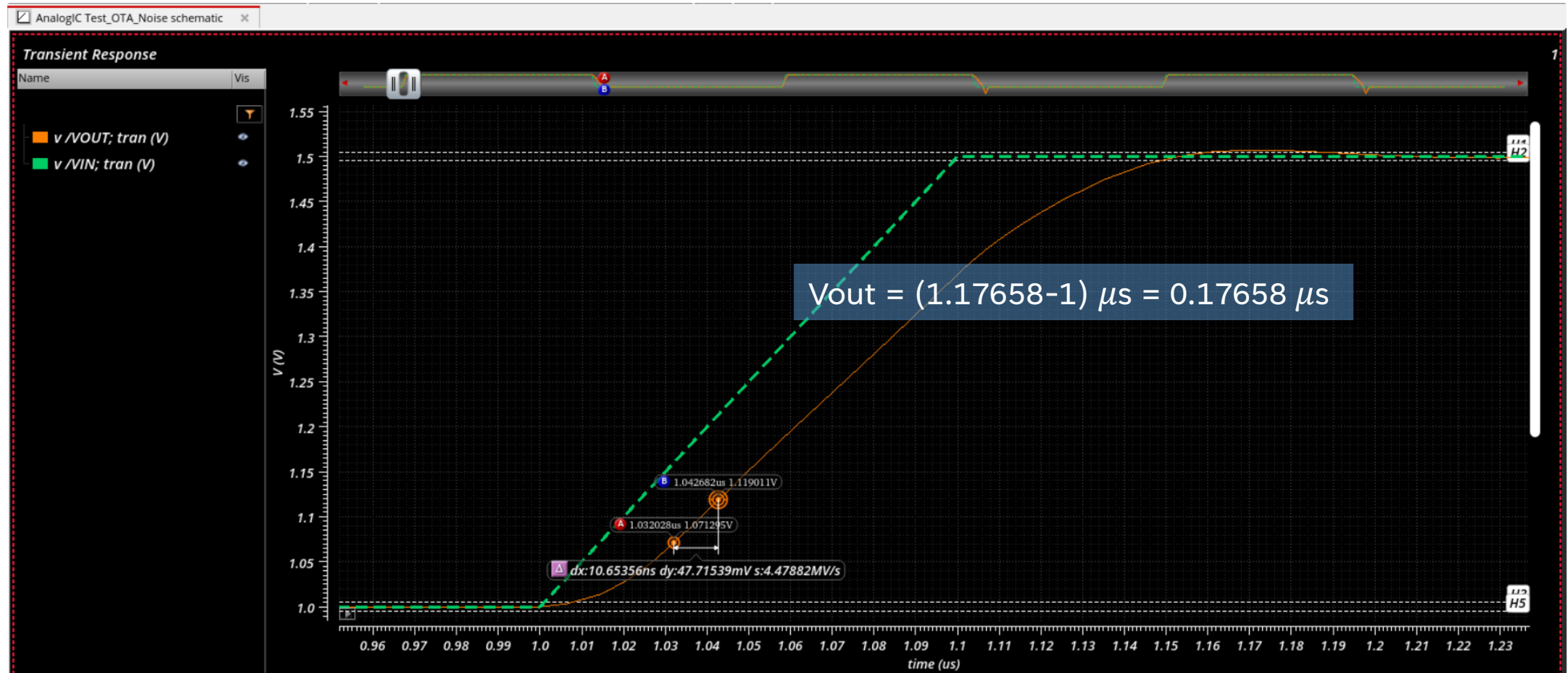
$$\text{TOTAL INTEGRATED INPUT-REFERRED NOISE} \leq 220 \mu V_{rms}$$



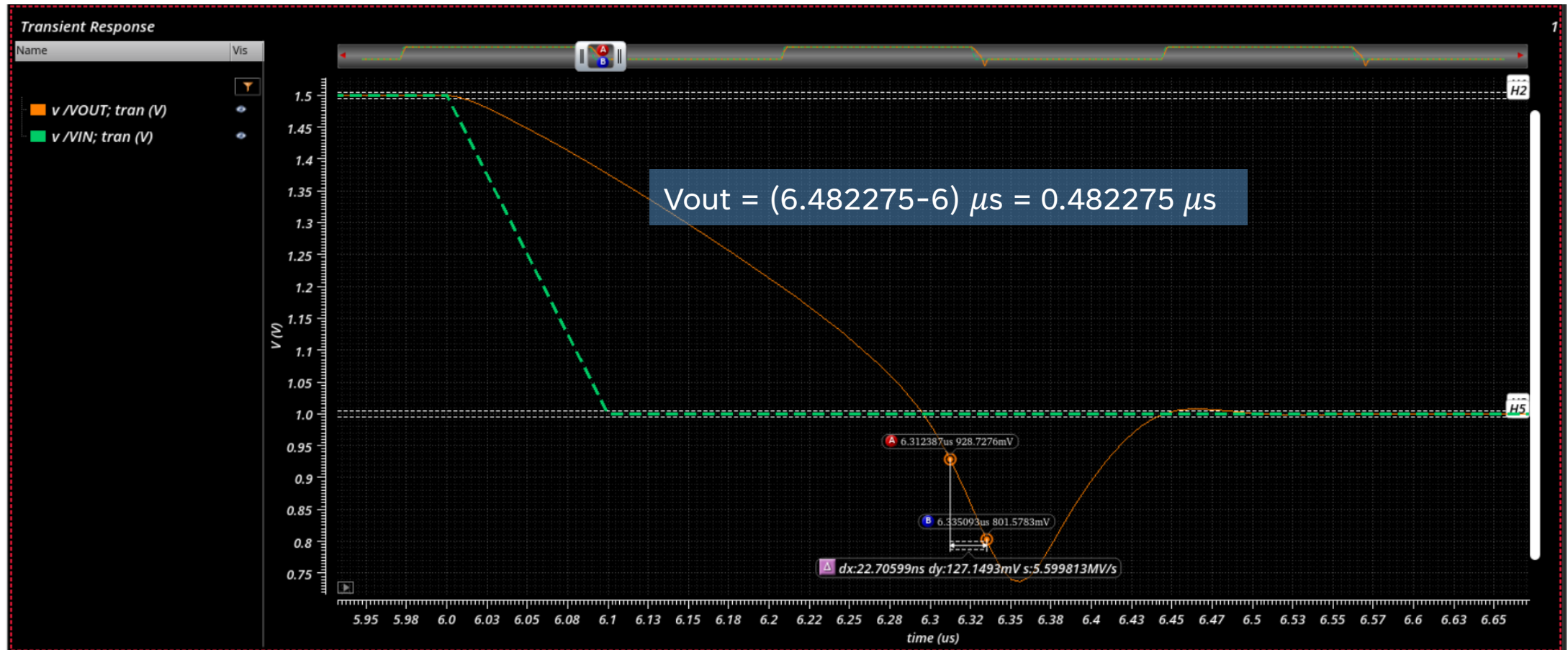
1 % SETTling TIME < 2 μ S WITH 0.5V STEP INPUT AND 1V DC OFFSET



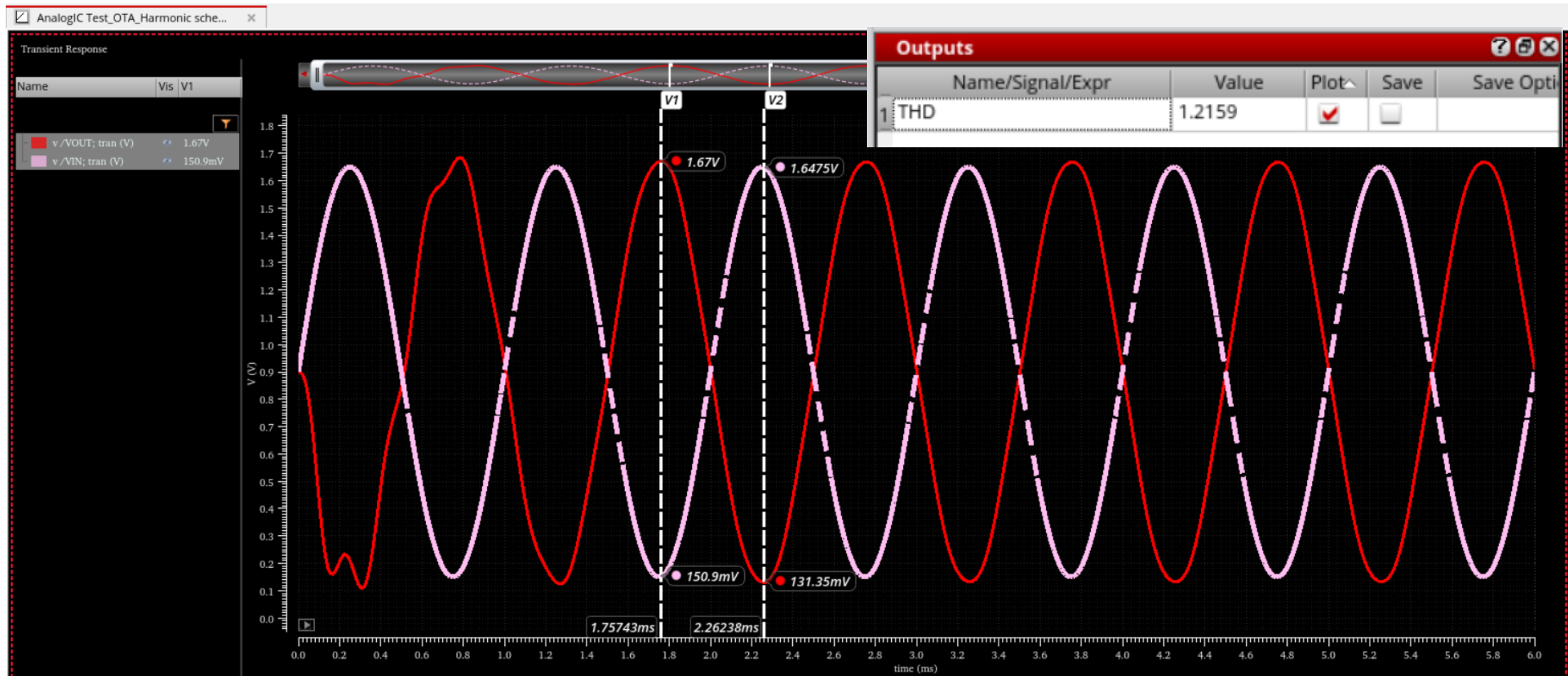
1 % SETTLING TIME < 2 μ S WITH 0.5V STEP INPUT AND 1V DC OFFSET



1 % SETTLING TIME < 2 μ S WITH 0.5V STEP INPUT AND 1V DC OFFSET



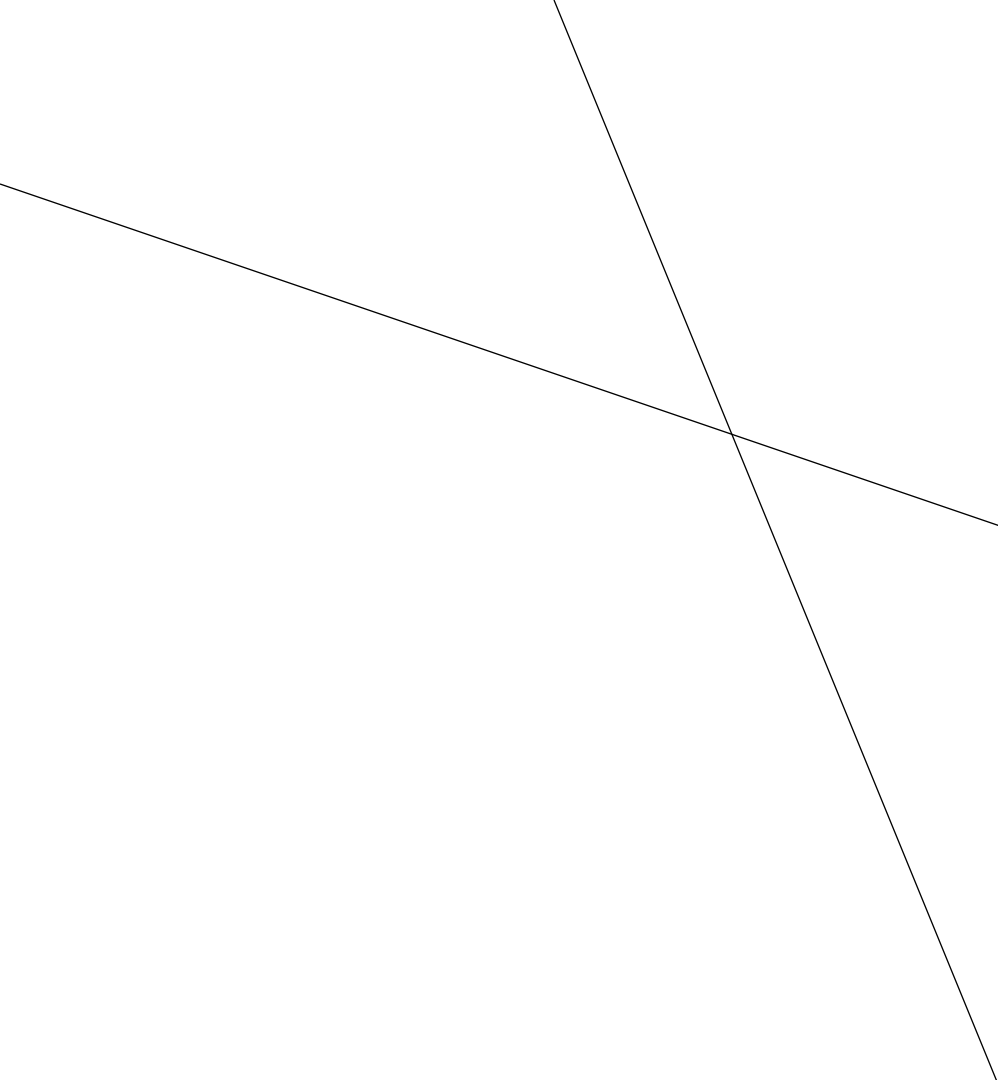
THD OF $V_{out} \leq 0.2 \%$ WHEN $V_{out} = 1.5 V_{pp}$



SUMMARY

SUMMARIZE TABLE

Performance	Result	Achieve Spec.
DC Gain	86.30 dB	✓
Output Swing	1.46 V	✗
Unity Gain	7.78 MHz	✓
Phase Margin	60.6 degree	✓
ICMR	1.42 V	✓
Noise	215.75 μ s	✓
Settling time	0.64 μ s	✓
THD	1.216 %	✗



SUMMARY

Almost all Specification are met excluded Output Swing which get 1.46 V and total harmonic distortion is 1.22 %

To adjust circuit, overdrive voltage of stage 2 must be decreased, and voltage swing should be 1.5 V in order to make lesser THD (No clipping)

A series of white, overlapping geometric lines and polygons on a black background, located on the left side of the slide.

THANK YOU

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